SNP Filters: Identity Analyses

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2026-08-20

Contents

Filtering the VCF

The raw VCF file is quite large (over 8Gb). We want to subset this large dataset to a set of markers that will be maximally informative for identity analyses.

The basic steps involved in the filters below include:

0. Cleaning the raw VCF - remove duplicated lines, half-missing genotype calls, indels
1. Quality filter at the locus level - drop the SNP locus if quality < 30
2. Maximum depth filter at the locus level - drop SNP loci if total sequencing depth is $> 2\sigma$ above mean total depth
3. Min depth filter at the genotype level - don't call genotypes with < 10 reads
4. Missing data filter - drop SNPs missing genotypes for $> 50\%$ of samples
5. Polymorphism filter - only retain loci that have exactly two alleles
6. Minor allele frequency filter - keep sites with MAF > 0.2
7. Avoiding linkage - prune markers based on LD within 500kb windows ($r^2 < 0.8$)

After all of these filters, the file `LGC_SCR.vcf` will contain SNP genotypes for samples processed by LGC / Rapid Genomics. In the next step, we will combine these with a smaller number of loci genotyped via GT-seq (by GTSeek). We will come back to the raw VCF used here to extract genotypes for loci from the GT-seq panel that did not pass the filters used in creating `LGC_SCR.vcf` (due to minor allele frequencies, missing data levels, etc.). We will also apply individual-level filters for missingness in the next step after creating the full, combined vcf file containing all samples and loci.

Step 0: Clean the raw VCF

First, copy the files and create some summaries of average depth, per-individual missing data, and per-genotype depth.

```
cd /scratch/jdrobins/CGA_SCR/snpfiltering

cp ../../LGC_BlackBears/UGA_173101/Delivery-20250620_merged/VCFs/UGA_173101_SNPs_Raw.vcf ./

#Use bcftools norm to remove duplicate SNPs
bcftools norm -d snps UGA_173101_SNPs_Raw.vcf -o filt0uniq.vcf

#Use sed to replace the half-missing calls (./0 and ./1) with missing data (./.)
grep ^# filt0uniq.vcf > filt0.vcf
grep -v ^# filt0uniq.vcf | sed -e 's/\.\/[0,1]/\.\/\./g' | sed -e 's/[0,1]\/\./\.\/\./g' >> filt0.vcf

#Use vcftools to remove indels
vcftools --vcf filt0.vcf --remove-indels --recode --out filt0

#make .idepth, .imiss, and .gdepth (this one is bigger) files
vcftools --vcf filt0.recode.vcf --out raw --depth
vcftools --vcf filt0.recode.vcf --out raw --missing-indv
vcftools --vcf filt0.recode.vcf --out raw --geno-depth

#How many SNPs do we start with?
grep -v ^# filt0.recode.vcf | wc -l
```

```
Lines   total/split/joined/realigned/mismatch_removed/dup_removed/skipped:  319680/0/0/0/0/5871/0

VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf filt0.vcf
    --out filt0
    --recode
    --remove-indels

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference all
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate all
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
```

```
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
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Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 827 out of 827 Individuals
Outputting VCF file...
After filtering, kept 312565 out of a possible 313809 Sites
Run Time = 996.00 seconds

VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf filt0.recode.vcf
    --depth
    --out raw

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference all
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate all
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
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Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
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Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
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Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 827 out of 827 Individuals
Outputting Mean Depth by Individual
After filtering, kept 312565 out of a possible 312565 Sites
Run Time = 43.00 seconds

VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf filt0.recode.vcf
    --missing-indv
```

```
    --out raw

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
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Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qual
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Like
After filtering, kept 827 out of 827 Individuals
Outputting Individual Missingness
After filtering, kept 312565 out of a possible 312565 Sites
Run Time = 39.00 seconds

VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf filt0.recode.vcf
    --geno-depth
    --out raw

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qual
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Like
After filtering, kept 827 out of 827 Individuals
```
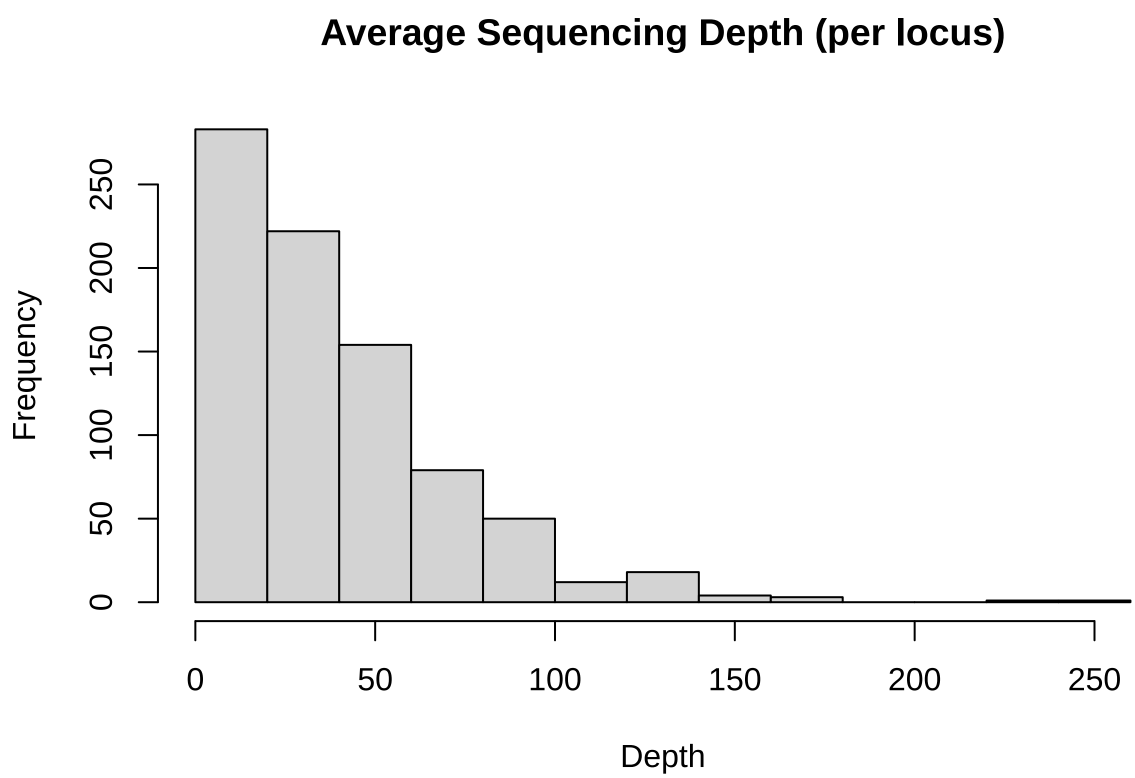
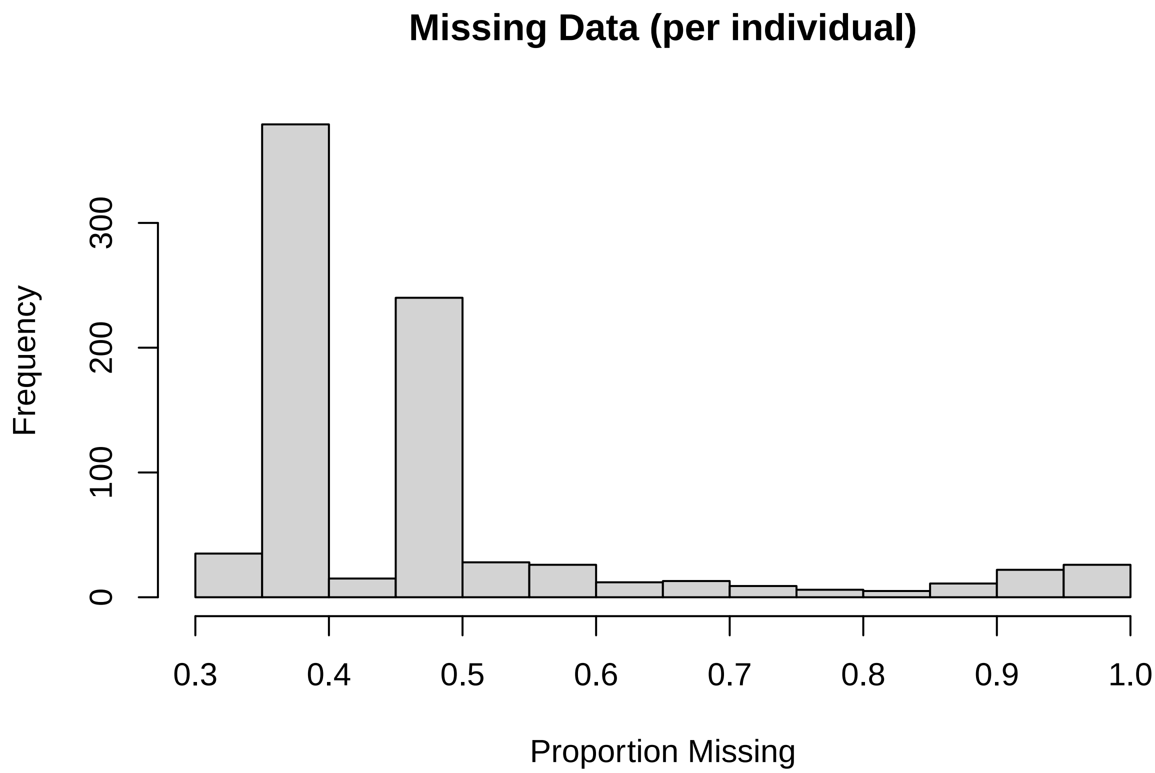

```
Outputting Depth for Each Genotype
After filtering, kept 312565 out of a possible 312565 Sites
Run Time = 84.00 seconds
312565
```

Plot some summaries of depth and missingness…

```r
depth <- read.table("raw.idepth", header = TRUE)
miss <- read.table("raw.imiss", header = TRUE)

hist(depth$MEAN_DEPTH, main = "Average Sequencing Depth (per locus)",
     xlab = "Depth")
```




```r
hist(miss$F_MISS, main = "Missing Data (per individual)",
     xlab = "Proportion Missing")
```




```r
#Quantiles of depth and missingness
quantile(depth$MEAN_DEPTH)
```

```
##        0%       25%       50%       75%      100%
##   1.70952  14.48280  31.35840  52.97025 255.97600
```

```r
quantile(miss$F_MISS)
```

```
##        0%       25%       50%       75%      100%
## 0.3491590 0.3525955 0.3999840 0.4762450 0.9961320
```

Step 1: Genotype Quality

The first filter we apply is a genotype quality filter (`--minQ`).

Prior to these filters, we have 3.12565×10^5 loci in the VCF…

```bash
#Filter sites
# min site quality of 30
vcftools --vcf filt0.recode.vcf --out filt1 \
          --minQ 30 --remove-filtered-all --recode
```

```
VCFtools - 0.1.17
```

```
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf filt0.recode.vcf
    --minQ 30
    --out filt1
    --recode
    --remove-filtered-all

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference all
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate all
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
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Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 827 out of 827 Individuals
Outputting VCF file...
After filtering, kept 96917 out of a possible 312565 Sites
Run Time = 323.00 seconds
```

Step 2: Maximum Depth

Next, we remove sites that are sequenced far more deeply than other markers. These may be loci where paralagous regions of the genome have been combined. We would expect markers in such regions to show elevated diversity and deviate from Hardy-Weinberg equilibrium. There are other ways to remove these (see McKinney et al. 2017), but we'll use a simple maximum (total) depth filter.

Prior to these filters, we have 9.6917×10^4 SNPs in the dataset…

```
#Get summaries of mean depth per site (for max depth filter)
vcftools --vcf filt1.recode.vcf --site-depth --out filt1
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf filt1.recode.vcf
    --out filt1
    --site-depth
```

```
Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference all
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate all
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
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Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 827 out of 827 Individuals
Outputting Depth for Each Site
After filtering, kept 96917 out of a possible 96917 Sites
Run Time = 14.00 seconds
```

Now in R, we can use this `filt1.ldepth` file to calculate a mean total depth (summed across individuals)…

```r
ldep <- read.table("filt1.ldepth", header = TRUE)

mean(ldep$SUM_DEPTH)
```

```
## [1] 20672.05
```

```r
sd(ldep$SUM_DEPTH)
```

```
## [1] 14356.02
```

```r
#This will be our threshold...
# nothing more than 2 sd above the mean...
cap <- mean(ldep$SUM_DEPTH) + (2*sd(ldep$SUM_DEPTH))
cap
```

```
## [1] 49384.1
```

```r
keeper.sites <- ldep[which(ldep$SUM_DEPTH <= cap),c(1,2)]
write.table(keeper.sites, file="keepersites.txt",
            quote=FALSE, row.names = FALSE, col.names = FALSE)
```

Now apply this filter…

```
#Then remove loci based on this threshold...
# only keep the SNPs in the list created above
vcftools --vcf filt1.recode.vcf --out filt2 \
         --positions keepersites.txt \
         --remove-filtered-all --recode
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
    --vcf filt1.recode.vcf
    --out filt2
    --positions keepersites.txt
    --recode
    --remove-filtered-all

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
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Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qual
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 827 out of 827 Individuals
Outputting VCF file...
After filtering, kept 92692 out of a possible 96917 Sites
Run Time = 311.00 seconds
```

Step 3: Genotype Depth

In the next filter, we remove genotype calls that are based on fewer than 10 reads. We set this threshold to maximize our confidence in individual calls. This is very important for identity analyses where genotyping errors lead to mismatches and may inflate the number of unique individuals detected in our project.

Prior to these filters, we have 9.2692×10^4 loci in the VCF…

```
#Min depth for genotype calls is 10 reads
vcftools --vcf filt2.recode.vcf --out filt3 \
         --minDP 10 --remove-filtered-all --recode
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
	--vcf filt2.recode.vcf
	--minDP 10
	--out filt3
	--recode
	--remove-filtered-all

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
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Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
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Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qual
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Like
After filtering, kept 827 out of 827 Individuals
Outputting VCF file...
After filtering, kept 92692 out of a possible 92692 Sites
Run Time = 310.00 seconds
```

Step 4: Missing Data (per locus)

We also remove loci that are missing data for more than 50% of samples. This reduces the mean proportion of missing genotypes across samples substantially, compared to more relaxed missing data filters. Note that the `--max-missing` argument to `vcftools` is defined such that "0 allows sites that are completely missing and 1 indicates no missing data allowed" (see the `vcftools` manual).

Prior to these filters, we have 9.2692×10^4 loci in the VCF...

```
#Max missing is 30% of samples (per locus)
vcftools --vcf filt3.recode.vcf --out filt4 \
            --max-missing 0.5 --remove-filtered-all --recode
```

```
VCFtools - 0.1.17
(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:
```

```
    --vcf filt3.recode.vcf
    --max-missing 0.5
    --out filt4
    --recode
    --remove-filtered-all

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qual
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Like
After filtering, kept 827 out of 827 Individuals
Outputting VCF file...
After filtering, kept 33490 out of a possible 92692 Sites
Run Time = 124.00 seconds
```

Step 5: Polymorphism filter

Now we want to apply a filter to remove sites that have more or less than 2 alleles. The built in `--min-alleles` and `--max-alleles` arguments in `vcftools` do not work the way we would want them to. They ignore variation present in the VCF at that stage, and instead base the number of alleles at a marker on the original calls (*i.e.*, the number of alleles implied by the `REF` and `ALT` columns of the file). This unnecessarily removes "multiallelic" sites that become biallelic after depth filters, which amounted to several thousand SNP loci in our dataset. Instead, we'll use the `--positions` argument to specify a keeper list of SNP loci where there are two alleles.

First, generate an allele frequency summary for the vcf using `vcftools`...

```
#Output a file of allele frequencies at each SNP
vcftools --vcf filt4.recode.vcf --out filt4 \
         --freq
```

```
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Parameters as interpreted:
    --vcf filt4.recode.vcf
    --freq
```

```
    --out filt4
```

```
Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference all
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate all
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balanc
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balanc
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qua
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Lik
After filtering, kept 827 out of 827 Individuals
Outputting Frequency Statistics...
After filtering, kept 33490 out of a possible 33490 Sites
Run Time = 5.00 seconds
```

Now read this file into R and identify sites that have exactly two alleles in the dataset...

```
snpfreq <- read.table("filt4.frq", skip = 1, fill=TRUE)
colnames(snpfreq) <- c("CHROM","POS","N_ALLELES","N_CHR", "A1","A2","A3")
biallelic <- snpfreq[which(snpfreq$N_ALLELES == 2),c(1,2)]

fixed.sites <- c(grep(":1", biallelic$A1),
                 grep(":1", biallelic$A2),
                 grep(":1", biallelic$A3))
fixed.sites
```

```
## integer(0)
```

```
biallelic[fixed.sites,]
```

```
## [1] CHROM POS
## <0 rows> (or 0-length row.names)
```

```
if(length(fixed.sites) > 0) {
  biallelic <- biallelic[-fixed.sites,c(1,2)]
}

nrow(biallelic)
```

```
## [1] 26134
```

```r
write.table(biallelic, file = "biallelic.txt",
            quote = FALSE, row.names = FALSE, col.names = FALSE)
```

Now retain only SNP loci in this list . . .

```
#Only retain SNPs in the file created above
vcftools --vcf filt4.recode.vcf --out filt5 \
        --positions biallelic.txt \
        --remove-filtered-all --recode
```

```
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Parameters as interpreted:
    --vcf filt4.recode.vcf
    --out filt5
    --positions biallelic.txt
    --recode
    --remove-filtered-all

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qual
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Like
After filtering, kept 827 out of 827 Individuals
Outputting VCF file...
After filtering, kept 26134 out of a possible 33490 Sites
Run Time = 88.00 seconds
```

Step 6: Minor Allele Frequency

The power for individual identification is maximized in biallelic loci with equal allele frequencies ($p = q = 0.5$). Thus, loci with high minor allele frequencies (MAF) will be the most useful for identifying recaptures. Here, we remove sites with MAF < 0.2. Note that the GT-seq panel that we're using has some low frequency SNPs included. In subsequent steps (*i.e.*, `4_combineGTSeek`), we'll come back to the raw VCF and extract genotypes for markers that are excluded here to retain loci from the GT-seq panel. However, the bulk of our dataset will have high MAF, maximizing the information content of each genotyped SNP locus.

Prior to these filters, we have 2.6134×10^4 loci in the VCF…

```
#MAF filter - remove SNPs with minor allele frequency < 0.2
vcftools --vcf filt5.recode.vcf --out filt6 \
         --maf 0.2 --remove-filtered-all --recode
```

```
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Parameters as interpreted:
    --vcf filt5.recode.vcf
    --maf 0.2
    --out filt6
    --recode
    --remove-filtered-all

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference allo
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate allo
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
Warning: Expected at least 2 parts in FORMAT entry: ID=GQ,Number=1,Type=Float,Description="Genotype Qual
Warning: Expected at least 2 parts in FORMAT entry: ID=GL,Number=G,Type=Float,Description="Genotype Like
After filtering, kept 827 out of 827 Individuals
Outputting VCF file...
After filtering, kept 1390 out of a possible 26134 Sites
Run Time = 8.00 seconds
```

Step 7: Pruning based on Linkage Disequilibrium (LD)

Finally, tightly linked loci do not provide independent information on individual identity, as they are inherited together. Thus, we need to subsample the SNP loci to avoid markers that are tightly linked. Below, we use PLINK2 to prune markers that exhibit high LD within 500 kb windows ($r^2 < 0.8$).

Prior to these filters, we have 1390 loci in the VCF…

```
#Create pruning files (LDprune prefix) with sets of markers
# to remove (prune.out suffix) and keep (prune.in suffix)
plink2 --vcf filt6.recode.vcf --allow-extra-chr \
        --indep-pairwise 500 kb 1 0.8 \
```

```
        --set-all-var-ids @:# --out LDprune --silent

#Apply the prune.in file to keep only loci that pass the LD filter
plink2 --vcf filt6.recode.vcf --set-all-var-ids @:# \
        --allow-extra-chr --extract LDprune.prune.in \
        --recode vcf-4.2 --out LGC_SCR --silent

#Extract allele frequencies for these SNPs to calculate p_{ID}
plink2 --vcf LGC_SCR.vcf --allow-extra-chr --freq --out LGC_SCR --silent

#How many loci?
grep -v ^# LGC_SCR.vcf | wc -l
```

```
1144
```

After all filters are applied, we have 1144 loci in the VCF. These will be combined with SNPs genotyped via GT-seq to form the dataset we use for the recapture analyses.

Summarize Dataset

Now we'll i) summarize the extent of missing data in our filtered VCF and ii) record the number of genotyped markers for each sample in our metadata file.

Missingness

After combining SNP genotypes from LGC / RapidGenomics with those from GTSeek, we will impose a filter to drop samples based on missing data. As a preliminary assessment of the number of samples we're likely to lose, we now evaluate the proportion of missing data in the LGC SNP set.

First, create files with summaries of the missing data levels by locus and by sample.

```
#Missing data by locus
vcftools --vcf LGC_SCR.vcf --missing-site --out LGC_SCR

#Missing data by sample
vcftools --vcf LGC_SCR.vcf --missing-indv --out LGC_SCR
```

```
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Parameters as interpreted:
    --vcf LGC_SCR.vcf
    --out LGC_SCR
    --missing-site

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference all
```

```
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
After filtering, kept 827 out of 827 Individuals
Outputting Site Missingness
After filtering, kept 1144 out of a possible 1144 Sites
Run Time = 1.00 seconds

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Parameters as interpreted:
    --vcf LGC_SCR.vcf
    --missing-indv
    --out LGC_SCR

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated allel
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extende
After filtering, kept 827 out of 827 Individuals
Outputting Individual Missingness
After filtering, kept 1144 out of a possible 1144 Sites
Run Time = 0.00 seconds
```

By locus

Read this into R and check the mean and range of the proportion of genotypes missing across loci.

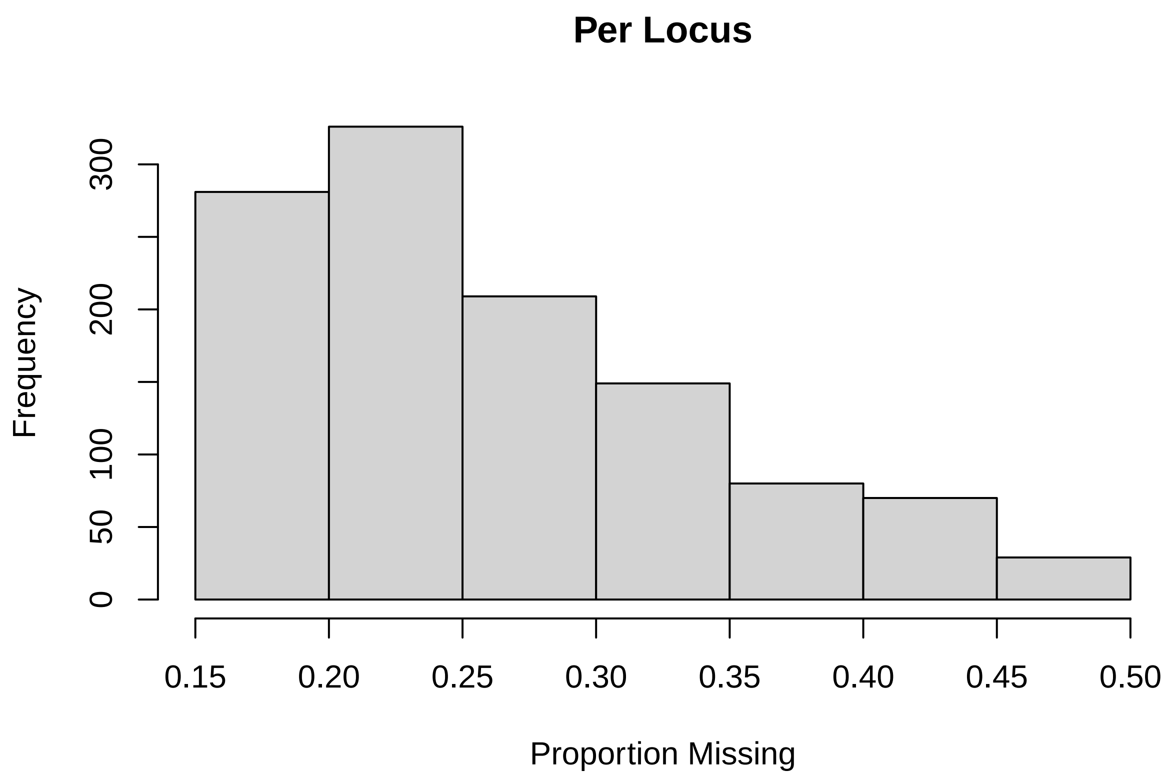
```r
missLocus <- read.table("LGC_SCR.lmiss", header = TRUE)
mean(missLocus$F_MISS)
```

```
## [1] 0.264709
```

```r
range(missLocus$F_MISS)
```

```
## [1] 0.155985 0.495768
```

```r
hist(missLocus$F_MISS,
     xlab = "Proportion Missing", ylab = "Frequency", main = "Per Locus")
```



By sample

Now the proportion of missing data for each sample.

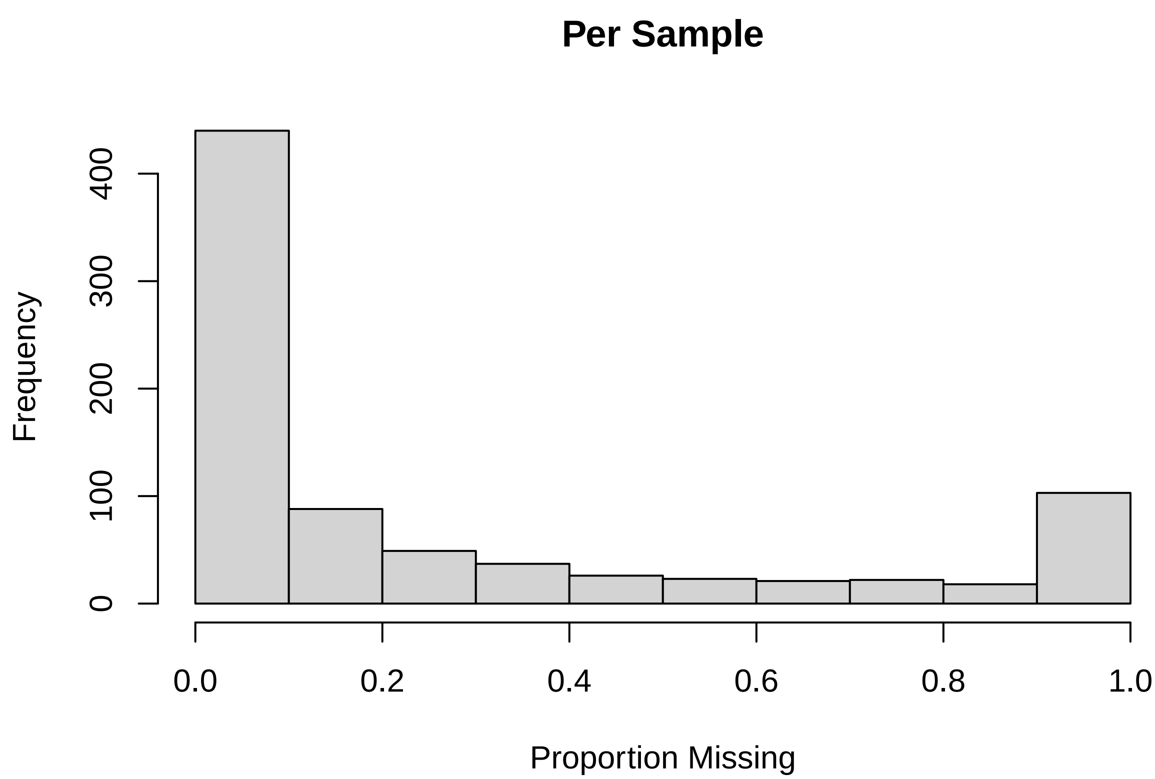
```r
missSamp <- read.table("LGC_SCR.imiss", header = TRUE)
mean(missSamp$F_MISS)
```

```
## [1] 0.264709
```

```r
range(missSamp$F_MISS)
```

```
## [1] 0 1
```

```r
hist(missSamp$F_MISS,
      xlab = "Proportion Missing", ylab = "Frequency", main = "Per Sample")
```




```r
#How many samples do we lose with various missing data filters...
nrow(missSamp) #total samples in the VCF
```

```
## [1] 827
```

```r
#Lost if keeping samples...
sum(missSamp$F_MISS >= 0.9) # with <90% missing
```

```
## [1] 103
```

```r
sum(missSamp$F_MISS >= 0.8) # with <80% missing
```

```
## [1] 121
```

```r
sum(missSamp$F_MISS >= 0.7) # with <70% missing
```

```
## [1] 143
```

```r
sum(missSamp$F_MISS >= 0.6) # with <60% missing
```

```
## [1] 164
```

```r
sum(missSamp$F_MISS >= 0.5) # with <50% missing
```

```
## [1] 188
```

```r
sum(missSamp$F_MISS >= 0.4) # with <40% missing
```

```
## [1] 213
```

```r
sum(missSamp$F_MISS >= 0.3) # with <30% missing
```

```
## [1] 250
```

```r
sum(missSamp$F_MISS >= 0.2) # with <20% missing
```

```
## [1] 299
```

```r
sum(missSamp$F_MISS >= 0.1) # with <10% missing
```

```
## [1] 387
```

Given that the genotypes we are working with come from non-invasive (hair) samples, higher thresholds for successful genotyping are unrealistic.

Record metadata

Finally, add information on the number of SNPs genotyped in each sample to our sample metadata file.

```r
smd <- read.csv("../sampleinfo/sampleMetadata_basic.csv", header = TRUE)

#Add the number of genotyped SNPs in the filtered VCF to sampleMetaData
smd$nSNPgt <- rep(NA, nrow(smd))
for(i in 1:nrow(smd)) {
  tmp <- which(missSamp$INDV == smd$rgID[i])
  smd$nSNPgt[i] <- missSamp$N_DATA[tmp]-missSamp$N_MISS[tmp]
  rm(tmp)
}
write.csv(smd, "sampleMetadata_nSNPgt.csv", quote = FALSE, row.names=FALSE)
```